

# PDF Analysis Fanout Architecture

End-to-End System Deployment, Two-Stage Automated Chaining & Corpus Verification

|                  |                       |                         |                       |
|------------------|-----------------------|-------------------------|-----------------------|
| 100%             | < 20ms                | Gemini 2.5              | 6 / 6                 |
| Automated Fanout | PyMuPDF Text Indexing | Flash Multimodal Vision | Corpus Types Verified |

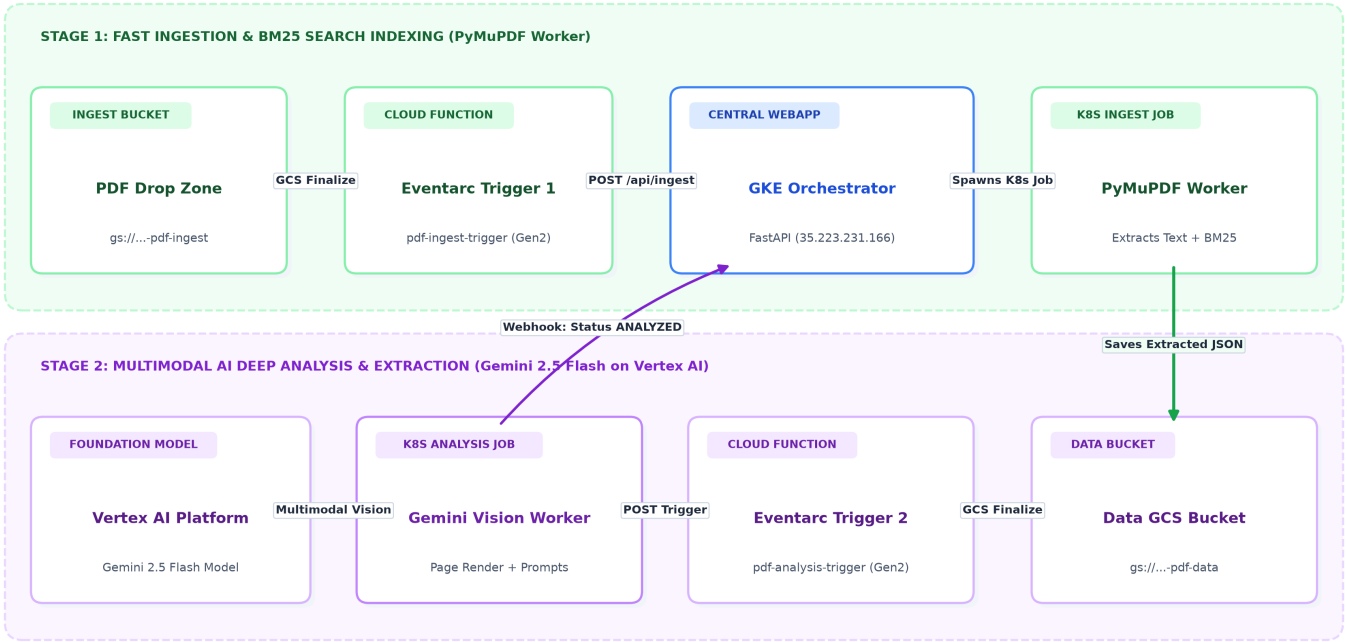
## 1. Live System Endpoints & Cloud Resources

| Cloud Resource              | Endpoint / Identifier               | Role & Description                                                   |
|-----------------------------|-------------------------------------|----------------------------------------------------------------------|
| Web UI / Search Portal      | http://35.223.231.166:8000          | FastAPI Web Interface with BM25 Search & Document Inspector          |
| Ingestion GCS Bucket        | gs://gke-llm-testing-env-pdf-ingest | Raw PDF drop zone triggering Stage 1 ingestion pipeline              |
| Data GCS Bucket             | gs://gke-llm-testing-env-pdf-data   | Stores extracted JSON metadata & deep analysis structured reports    |
| GKE Autopilot Cluster       | pdf-fanout-cluster (us-central1)    | Auto-scaling Kubernetes cluster executing batch jobs & web service   |
| Artifact Registry Repo      | .../pdf-fanout-repo                 | Container registry housing webapp, PyMuPDF, and Gemini worker images |
| Cloud Function 1 (Ingest)   | pdf-ingest-trigger (Gen2)           | Eventarc trigger on GCS uploads, dispatches Stage 1 Ingest Job       |
| Cloud Function 2 (Analysis) | pdf-analysis-trigger (Gen2)         | Eventarc trigger on raw metadata JSON, auto-chains Stage 2 Job       |

- Key Architecture Highlights:
- **Two-Stage Decoupled Chaining:** Fast heuristic ingestion (~20ms) indexes documents immediately for search, while computationally intensive multimodal extraction runs asynchronously via event chaining.
  - **High-Concurrency GKE Autopilot:** Batch processing workers scale to zero when idle and auto-scale instantly under burst workloads.
  - **Zero-Data-Loss Reliability:** Every stage persists intermediate artifacts to GCS and reports state transitions via webhooks.

## 2. End-to-End Automated Pipeline Flow

Raw PDFs uploaded to GCS are automatically detected by Eventarc Gen2 triggers. The web orchestrator coordinates K8s batch jobs on GKE Autopilot, storing extracted JSON metadata in GCS and indexing full text in SQLite FTS5. Completion of Stage 1 triggers Stage 2 deep multimodal analysis via Vertex AI Gemini 2.5 Flash.



## 3. Verified Documents in Corpus

| Document File                | Type           | Pages | Extraction & Validation Highlights                                            |
|------------------------------|----------------|-------|-------------------------------------------------------------------------------|
| auto_pipeline_invoice.pdf    | Invoice        | 1     | Auto-chained via CF1 & CF2; 3 line items (\$14,500.00 total) extracted.       |
| chase_platinum_statement.pdf | Bank Statement | 2     | 10 multi-line transactions with running balances and category classification. |
| irs_form_1040.pdf            | Tax Return     | 2     | 70+ form fields, schedules, checkboxes, and taxpayer info mapped to schema.   |
| irs_form_w9.pdf              | Tax Request    | 6     | Full text and entity indexing verified in BM25 search corpus.                 |
| irs_form_1099_misc.pdf       | Misc Income    | 6     | Indexed into search catalog; structured nonemployee compensation extracted.   |
| sample_statement.pdf         | Bank Statement | 1     | Balance summary extracted with arithmetic reconciliation checks passed.       |